

Model Question Paper-I with effect from 2025-26

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1BMATC201

Second Semester B.E./B.Tech. Degree Examination

Differential Calculus and Numerical Methods

TIME: 03Hours

Max.Marks:100

Note: 1. Answer any FIVE full questions, choosing at least ONE question from each MODULE.

2. VTU Formula Hand Book is Permitted

3. M: Marks, L: Bloom's level, C: Course outcomes.

Module-1			M	L	COS
Q 1.	a	Evaluate $\int_{-1}^1 \int_0^z \int_{x-z}^{x+z} (x + y + z) dy dx dz$	6	L2	CO-1
	b	Change the order of the integration and hence evaluate $\int_0^1 \int_0^{\sqrt{1-x^2}} y^2 dx dy$	7	L2	CO-1
	c	Show that the area between the parabola $y^2 = 4ax$, $x^2 = 4ay$ is $\frac{16}{3}a^2$	7	L2	CO-1
OR					
Q 2.	a	Find the volume generated by the revolution of the cardioid $r = a(1 + \cos\theta)$	6	L2	CO-1
	b	Prove that $\beta(m, n) = \frac{[m] [n]}{[m+n]}$	7	L2	CO-1
	c	Prove that $\int_0^{\frac{\pi}{2}} \frac{d\theta}{\sqrt{\sin\theta}} \times \int_0^{\frac{\pi}{2}} \sqrt{\sin\theta} d\theta = \pi$	7	L2	CO-1
Module-2					
Q 3.	a	Form the partial differential equation by eliminating arbitrary constant from. $\log(az - 1) = x + ay + b$	6	L2	CO-1
	b	Form the partial differential equation by eliminating arbitrary function from $z = f(x + ay) + g(x - ay)$	7	L2	CO-1
	c	Solve $\frac{\partial^2 u}{\partial x \partial y} + 9x^2 y^2 = \cos(2x - y)$ given that $u = 0$ when $y = 0$ and $\frac{\partial u}{\partial y} = 0$ when $x = 0$	7	L2	CO-1
OR					

Q 4.	a	Solve $\frac{\partial^2 z}{\partial x^2} - 6 \frac{\partial z}{\partial x} + 9z = 0$ given that $z=0$ and $\frac{\partial z}{\partial x} = e^y$ when $x=0$.	6	L2	CO-1
	b	Solve by the method of separation of variables $4 \frac{\partial u}{\partial x} + 3 \frac{\partial u}{\partial y} = u$	7	L2	CO-1
	c	Derive the one dimensional heat equation $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$	7	L2	CO-1
Module-3					
Q 5.	a	Find the directional derivative of $\phi = 4xz^3 - 3x^2y^2z$ at $(2, -1, 2)$ along $2i-3j+6k$	6	L2	CO-1
	b	Find $\text{Div}(\vec{F})$, $\text{Div}(\text{curl}(\vec{F}))$ for the vector $\vec{F} = y^2i + 2x^2yzj + x - 3y^2zk$ at the point $(2,1,1)$.	7	L2	CO-1
	c	If $\vec{A} = xz^3i - 2x^2yzj + 2yz^4k$. Find $\nabla \cdot A$, $\nabla \times A$ and $\nabla \cdot (\nabla \times A)$	7	L2	CO-1
OR					
Q 6.	a	Find the value of the constant “a” such that the vector field $\vec{F} = (axy - z^3)i + (a - 2)x^2j - (1 - a)xz^2k$ is irrotational and hence find scalar function ϕ such that $\vec{F} = \nabla\phi$	6	L2	CO-1
	b	If $\vec{F} = xy i + yz j + zx k$ evaluate $\int_c \vec{F} \cdot d\vec{r}$ where c is the curve represented by $x = t, y = t^2, z = t^3, -1 \leq t \leq 1$	7	L2	CO-1
	c	Find the area between the curves $4y = x^2$, and $y^2 = 4x$ with the help of the Green's theorem in a plane.	7	L2	CO-1
Module-4					
Q 7.	a	Find the real root of the equation $xe^x - \cos x = 0$ by method of false position.	6	L2	CO-2
	b	Using Newton Raphson method to find the real root of the equation $x \sin x + \cos x = 0$ near $x = \pi$, correct to 4 decimal places.	7	L2	CO-2
	c	The area of a circle(A) corresponding to diameter(D) is given below: D 80 85 90 95 100 A 5026 5674 6362 7088 7854 find the area corresponding to diameter 105 using an appropriate interpolation formula.	7	L2	CO-2
OR					
Q 8.	a	Evaluate $\int_4^{5.2} \log x \, dx$ by 6 equal parts by Simpson's three eight rule	6	L2	CO-2
	b	Evaluate $\int_0^1 \frac{x \, dx}{1+x^2}$ taking into six equal parts. By using Simpson's $\frac{3}{8}$ rule.	7	L2	CO-2

	c	Use Lagraenge`s interpolation formula fit a polynomial for the data. <table><tr><td>x</td><td>0</td><td>1</td><td>3</td><td>4</td></tr><tr><td>y</td><td>-12</td><td>0</td><td>6</td><td>12</td></tr></table>	x	0	1	3	4	y	-12	0	6	12	7	L2	CO-2
x	0	1	3	4											
y	-12	0	6	12											
Module-5															
Q 9.	a	Find y at x=1.02 correct to five decimal places given $dy = (xy - 1)dx$ and y=2 at x=1 applying Taylor`s series method.	6	L2	CO-2										
	b	Use Modified Euler`s method to compute y(1.2) correct to five decimal places given that $\frac{dy}{dx} + \frac{y}{x} = \frac{1}{x^2}$ and $y = 1$ at $x = 1$, taking $h = 0.1$.	7	L2	CO-2										
	c	Use forth order Runge-kutta method to solve $(x + y)\frac{dy}{dx} = 1$, $y(0.4) = 1$, at $x=0.5$ correct to four decimal places.	7	L2	CO-2										
OR															
Q 10.	a	Solve the differential equation $5x\frac{dy}{dx} + y^2 - 2 = 0$ with $y(4) = 1$,initially $y(4.1)=1.0049$, $y(4.2)=1.0097$, $y(4.3)=1.0049$ using Milne`s method.	6	L2	CO-2										
	b	If $\frac{dy}{dx} = 2e^x - y$, $y(0)=2$, $y(0.1)=2.010$, $y(0.2)=2.040$, $y(0.3)=2.090$, find $y(0.4)$ correct to four decimal places by using Adams- Bashforth predictor method	7	L2	CO-2										
	c	Use forth order Runge-kutta method to solve $\frac{dy}{dx} = \frac{y-x}{(x+y)}$, $y(0) = 1$, at $x=0.1$ correct to four decimal places.	7	L2	CO-2										

CBCS SCHEME

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1	✓	5	2	5	✓	0	0	1
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1BMATC201

Second Semester B.E./B.Tech. Degree Examination, June/July 2026 Differential Calculus and Numerical Methods : CV Stream

Time: 3 hrs.

Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks, L: Bloom's level, C: Course outcomes.
3. VTU Formula Hand Book is permitted.*

Module - 1			M	L	C
Q.1	a.	Find $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx dy$ by changing to polar coordinates	6	L2	CO1
	b.	Find $\int_{-1}^1 \int_0^2 \int_{x-z}^{x+z} (x+y+z) dx dy dz$	7	L2	CO1
	c.	Show that $\beta(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)}$	7	L2	CO1
OR					
Q.2	a.	Change the order of integration and hence find $\int_0^1 \int_0^{\sqrt{1-x^2}} y^2 dy dx$	6	L2	CO1
	b.	Find the area between the parabolas $y^2 = 4ax$ and $x^2 = 4ay$ using double integral.	7	L2	CO1
	c.	Show that $\int_0^{\pi/2} \frac{d\theta}{\sqrt{\sin \theta}} \cdot \int_0^{\pi/2} \sqrt{\sin \theta} d\theta = \pi$	7	L2	CO1
Module - 2					
Q.3	a.	Form a partial differential equation of the family of sphere $(x-a)^2 + (y-b)^2 + z^2 = r^2$ by eliminating arbitrary constants.	6	L2	CO1
	b.	Solve $\frac{\partial^2 z}{\partial x^2} + z = 0$, given that, $z = e^y$ and $\frac{\partial z}{\partial x} = 1$ when $x = 0$	7	L2	CO1
	c.	Derive of one-dimensional heat equation.	7	L2	CO1
OR					
Q.4	a.	Form a partial differential equation by eliminating the arbitrary functions f and g from the relation $z = f(x+ay) + g(x-ay)$.	6	L2	CO1
	b.	Solve $\frac{\partial^2 z}{\partial x \partial y} = \sin x \sin y$ for which $\frac{\partial z}{\partial y} = -2 \sin y$, when $x = 0$ and $z = 0$ when $y = (2n+1)\frac{\pi}{2}$.	7	L2	CO1
	c.	Find the solution of $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 2(x+y)u$ by the method of separation of variables.	7	L2	CO1

Module - 3

Q.5	a.	Find the directional derivative of $\phi = x^2yz + 4xz^2$ along $2i - j - 2k$ at $(1, -2, -1)$	6	L2	CO1
	b.	Show that the vector field $\vec{F} = (x^2 - yz)i + (y^2 - zx)j + (z^2 - xy)k$ is irrotational and find its scalar potential.	7	L2	CO1
	c.	If $\vec{F} = xyi + yzj + zxk$, find $\int_C \vec{F} \cdot d\vec{r}$ where C is a curve represented by $x = t, y = t^2, z = t^3, -1 \leq t \leq 1$	7	L2	CO1

OR

Q.6	a.	If $\vec{F} = \nabla(x^3 + y^3 + z^3 - 3xyz)$ find $\nabla \cdot \vec{F}$ and $\nabla \times \vec{F}$	6	L2	CO1
	b.	Find the constants a, b, c so that vector function $\vec{F} = (x + 2y + az)i + (bx - 3y - z)j + (4x + cy + 2z)k$ is irrotational. Hence find scalar potential ϕ such that $\vec{F} = \nabla\phi$	7	L2	CO1
	c.	Using Green's theorem find $\int_C (xy - x^2) dx + x^2y dy$ where C is the closed curve formed by $y = 0, x = 1$ and $y = x$.	7	L2	CO1

Module - 4

Q.7	a.	Using Newton Raphson method find the real root of the equation $x^3 - 2x - 5 = 0$ correct to three decimal places.	6	L2	CO2												
	b.	From the following table find the number of students who scored less than 45 marks using Newton's forward interpolation formula <table border="1"><tr><td>Marks</td><td>30 - 40</td><td>40 - 50</td><td>50 - 60</td><td>60 - 70</td><td>70 - 80</td></tr><tr><td>No. of Students</td><td>31</td><td>42</td><td>51</td><td>35</td><td>31</td></tr></table>	Marks	30 - 40	40 - 50	50 - 60	60 - 70	70 - 80	No. of Students	31	42	51	35	31	7	L2	CO2
Marks	30 - 40	40 - 50	50 - 60	60 - 70	70 - 80												
No. of Students	31	42	51	35	31												
	c.	Find $\int_0^6 \frac{1}{1+x^2} dx$ using Simpson's $\left(\frac{1}{3}\right)^{rd}$ and Simpson's $\left(\frac{3}{8}\right)^{th}$	7	L2	CO2												

OR

Q.8	a.	Using Regula-Falsi method find the real root of the equation $x \log_{10} x = 1.2$ correct to 4 decimal places.	6	L2	CO2														
	b.	Find the interpolating polynomial $y = f(x)$ for the data given below using Newton's divided difference formula. <table border="1"><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>f(x)</td><td>3</td><td>2</td><td>7</td><td>24</td><td>59</td><td>118</td></tr></table>	x	0	1	2	3	4	5	f(x)	3	2	7	24	59	118	7	L2	CO2
x	0	1	2	3	4	5													
f(x)	3	2	7	24	59	118													

Q.8	c.	Using Lagrange's interpolation formula, find the interpolating polynomial that approximates the function described by the following data and hence find $f(3)$.	7	L2	CO2										
<table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>5</td></tr><tr><td>f(x)</td><td>2</td><td>3</td><td>12</td><td>147</td></tr></table>			x	0	1	2	5	f(x)	2	3	12	147			
x	0	1	2	5											
f(x)	2	3	12	147											

Module - 5

Q.9	a.	Employ Taylor series method to find an approximate solution to find y at $x = 0.1$ given $\frac{dy}{dx} = x - y^2$, $y(0) = 1$ by considering upto fourth degree term.	6	L2	CO2
	b.	Using Runge -Kutta method of fourth order, compute $y(0.2)$ given that $\frac{dy}{dx} = 3x + \frac{y}{2}$, $y(0) = 1$ by taking $h = 0.2$.	7	L2	CO2
	c.	If $\frac{dy}{dx} = 2e^x - y$ and $y(0) = 2$, $y(0.1) = 2.010$, $y(0.2) = 2.040$, $y(0.3) = 2.090$ find $y(0.4)$ corrected to 4 decimal places by using Milnes Predictor corrector formula.	7	L2	CO2

OR

Q.10	a.	Using Euler's modified method, find y at $x = 0.1$ given $\frac{dy}{dx} = \frac{y-x}{y+x}$, $y(0) = 1$ by taking $h = 0.1$	6	L2	CO2
	b.	Using Runge - Kutta method of fourth order, compute $y(1.1)$ given that $\frac{dy}{dx} = xy^{\frac{1}{3}}$, $y(1) = 1$ by taking $h = 0.1$	7	L2	CO2
	c.	Using Adams-Bashforth method, determine $y(0.4)$ given $\frac{dy}{dx} = x^2 + y^2$ and $y(0) = 1$, $y(0.1) = 1.1113$, $y(0.2) = 1.2507$, $y(0.3) = 1.426$.	7	L2	CO2
